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REVIEW ARTICLE OPEN ACCESS

The Prevalence of Insomnia Disorder in the General Population: A Meta-Analysis

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ABSTRACT

Insomnia disorder is a significant public health issue, but the prevalence estimates vary widely. We performed a meta-analysis aiming to pool prevalence rates in studies (1) carried out in the general population (2) using a true random sample (3) and using a diagnostic interview, DSM based self-report questions, or a questionnaire with a cut-off established against the DSM criteria. A literature search (in PubMed, Embase, APA PsycInfo) was performed up to April 2024. Two independent reviewers assessed title and abstracts ($n = 6732$), full-text manuscripts ($n = 621$) and extracted the data of the 47 included studies. Prevalence rates were pooled using a three-level hierarchical random-effects model, stratified by diagnosis type and adjusted for gender distribution and mean sample age. The pooled prevalence of all studies using an interview to establish the DSM criteria was 12.4% (95% CI: 9.0–16.8%), and of self-report questions assessing the DSM diagnosis 16.3% (95% CI: 11.3%–23.0%). There were 27 studies using different insomnia questionnaires with different cut-offs (prevalence range 7.5%–32.3%). The prevalences differed significantly across regions and high quality studies yielded a lower prevalence rates than lower quality studies. This meta-analysis confirms that insomnia is a common disorder with a prevalence of 12.4 as the most accurate estimate. It also shows the need for standardised ways of assessing insomnia. We think the golden standard is using standardised structured clinical interviews. However, if this is not feasible, we recommend using well validated questionnaires such as the Sleep Condition Indicator or the Insomnia Severity Index.

Trial Registration: PROSPERO CRD42023402745

1 | Introduction

Insomnia disorder, also known as chronic insomnia, is a significant public health issue. The definition of Insomnia Disorder has undergone important changes over time but is currently

defined by the Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association 2013) as having difficulties initiating or maintaining sleep, or unwanted early waking for at least 3 days a week for at least 3 months, despite adequate sleep opportunity. Additionally, the sleep

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disturbance should not solely be explained by another disorder, medication or substance use. Importantly, the sleep difficulties must also lead to clinically significant distress or impairment in social, occupational, educational, academic, behavioural, or other important areas of functioning (American Psychiatric Association 2013).

Chronic insomnia is associated with a range of adverse outcomes, including fatigue, cognitive impairments, mood disturbances, and diminished daytime functioning, all of which can significantly reduce quality of life (Grandner et al. 2023; Morin and Jarrin 2022; Olfson et al. 2018; Riemann et al. 2022; Spira et al. 2014). It is also linked to markedly reduced work productivity and high societal costs (Chaput et al. 2023; Daley et al. 2009). Furthermore, insomnia is associated to various other somatic and mental disorders such as cardiovascular disease, cancer, pain, depression and anxiety (Benz et al. 2023; Hertenstein et al. 2019; Luik et al. 2024). Consequently, understanding the prevalence of chronic insomnia in the general population is critical for public health and intervention planning.

There are numerous epidemiological studies reporting prevalence rates, but these estimates vary. Firstly, the differences might be explained by changes in the diagnostic criteria over time. For example, the DSM-III and DSM-IV distinguished primary and secondary insomnia, but this distinction has disappeared in the DSM-5. The DSM-5 also includes 'dissatisfaction' with sleep quantity or quality, which was not present in previous DSM versions. Furthermore, the frequency and duration of the symptoms have become more explicit in different DSM versions, with the duration of symptoms changing from 1 to 3 months (Substance abuse and mental health services 2016). Secondly, prevalence rates might vary because of their sampling method. Some studies are conducted in specific populations such as menopausal woman (Jia et al. 2024), during COVID-confinement (Yuan et al. 2022), or in people with psychiatric problems (Mijnster et al. 2024). Thirdly, the insomnia criteria might be assessed in different ways. Ideally, the diagnosis is established by a trained professional but using diagnostic interviews is labour intensive, and therefore many researchers use self-report questions. A validated example is the Sleep Condition Indicator (SCI) which consists of 8 questions and is based on the DSM-5 criteria (Espie et al. 2014). Other questionnaires which assess insomnia symptoms are also available but each one of them assesses slightly different things. The three most commonly used examples are the Insomnia Severity Index (ISI; Bastien et al. 2001), which is based on DSM-IV criteria and established a cut-off based on the sensitivity and specificity of detecting the diagnosis (Morin et al. 2011); the Athens Insomnia Scale (AIS; Soldatos et al. 2000), which is based on the ICD-10 criteria which are quite similar to the DSM-5 criteria but has no criteria for the frequency of poor nights nor for the duration of the symptoms; the Pittsburgh Sleep Quality Index (PSQI; Buysse et al. 1989), which assesses sleep problems but is not specific for insomnia. Next to these three there are a number of other insomnia questionnaires but these are used less frequently (Ali et al. 2020). And lastly, also questionnaires with only a few questions are commonly used. Examples of these questions are "do you have difficulty falling asleep at night?", "are you unable to get back

to sleep after an awakening?". These questions are almost never validated and oftentimes do not assess any daytime consequences.

More than 20 years ago, a landmark paper by Ohayon systematically reviewed all published prevalence studies based on DSM diagnosis (Ohayon 2002). He reported a prevalence rate of 6%. In 2022, Morin and Jarrin published another systematic review on insomnia prevalence. They observed that 10%–15% of the population reported insomnia with daytime symptoms, and 6%–10% suffered from insomnia according to the strict DSM/ICSD criteria. Many epidemiological studies have been conducted since, but to our knowledge no meta-analysis has been performed. Therefore, this paper aims to pool prevalence rates in studies (1) carried out in the general population (2) using a true random sample (3) and using (a) a DSM interview carried out in person (b) DSM based self-report questions or (c) a questionnaire with a cut-off established against the DSM criteria. We will explore the effects of the way insomnia is measured, as well as the effects of age, gender, and country.

2 | Methods

The conduct and reporting of this review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-statement (Page et al. 2021). The study is registered in the Prospero database (CRD42023402745). Checking the eligibility criteria and the quality of the studies was done by two persons independently as was data extraction. In case of disagreement at least one other team member was consulted.

2.1 | Search Strategy

To identify relevant publications, we conducted systematic searches in the bibliographic databases PubMed, Embase.com and Ebsco/APA PsycInfo from inception up to April 15, 2024, in collaboration with a medical information specialist. The following terms were used (including synonyms and closely related words) as index terms or free-text words: "Insomnia", "Prevalence", "Incidence", "Cohort", "Cross-sectional", "Longitudinal", "Follow-up", "Prospective", "Retrospective". The references of the identified articles were searched for relevant publications. Only manuscripts in English were included. Duplicate articles were excluded by a medical information specialist using Endnote X21.0.1 (Clarivatetm), following the Amsterdam Efficient Deduplication (AED)-method (Otten et al. 2019) and the Bramer-method (Bramer et al. 2016). The full search strategies for all databases can be found in the [Supporting Information](#).

2.2 | Inclusion Criteria

We included all studies with a population sample of (mostly) adults (18+), that reported a prevalence rate of insomnia. We allowed studies that included only men or women, studies performed in a certain age group, and studies performed in specific ethnic groups.

We distinguished three categories with respect to insomnia diagnosis: (1) established with help from a professional based on DSM-IV or DSM-5 criteria (2) patients' self-reported diagnosis based on questions that checked DSM-IV or DSM-5 criteria including day-time complaints, this also includes the SCI and (3) patients scoring above a validated cut-off on a insomnia questionnaire. We did not allow studies based on DSM-III criteria because daytime symptoms are not included in the diagnosis. Studies with the following validated questionnaires were included: the Insomnia Severity Index (ISI; Morin et al. 2011), the Athens Insomnia Scale (AIS; Soldatos et al. 2000), and the Holland Sleep Disorder Questionnaire (HSDQ; Kerkhof et al. 2013). We used the cut-offs as reported in the papers.

The sample had to be drawn from the general population and sampled at random. Convenience samples (e.g., asking for participation through social media) and snowballing methods were not allowed. We excluded studies in which the study population was selected on certain characteristics (e.g., receiving a certain type of care, having a certain disease, being menopausal) or when data was collected exclusively during the COVID pandemic. We also excluded papers on other sleep disorders than insomnia, papers that reported only mean scores and not proportions scoring above the cut-off, and papers that did not include data for pooling (editorials, letters etcetera).

2.3 | Data Extraction

The following data was extracted from all included articles: author, year of publication, country, the way insomnia was identified (DSM administered face-to-face/DSM using self-report questions/questionnaire), the instrument that was used to assess insomnia, the cut-off score used, the mean age of the sample, the number of people included in the total sample, and the number of people with insomnia. In addition to the overall prevalence rate, we also extracted prevalence rates for men and women separately when reported.

2.4 | Quality Assessment

We used the Joanna Briggs Institute Critical Appraisal Checklist for studies reporting prevalence data (Munn et al. 2015). This checklist has 9 items regarding the (1) sample frame (2) sampling method (3) sample size (4) sample description (5) sample coverage (6) identification methods (7) assessment instruments (8) statistical analysis (9) response rate and handling of low response rates. All items that were handled correctly were scored "yes", items handled incorrectly were scored "no" and items that were not reported were scored as "unclear". All studies with six or more positive answers were considered to have a low risk of bias and thus of high quality. The remaining studies were considered to have a moderate to high risk of bias and thus of lower quality.

2.5 | Analysis

All analyses were conducted in R (version 4.3.3) using the "metapsyTools" package; (Harrer et al. 2022), which imports

functionality of the "metafor", "meta", and "dmetar" packages (Viechtbauer 2010; Balduzzi et al. 2019; Harrer et al. 2021).

Prevalence was computed for all studies by dividing the number of participants with insomnia by the total number of participants. We then logit-transformed all prevalences and calculated (stabilised) sampling variances used for pooling. Studies were then pooled using a three-level hierarchical effects model (prevalence estimates nested in studies; Harrer et al. 2021), which was adjusted for differences in the gender distribution and mean age across studies. It is not useful to calculate an overall prevalence estimate across all studies, since these studies used different ways to establish the presence of insomnia. We therefore added stratification terms to our model, providing separate prevalence estimates depending on the method used to obtain the diagnosis. Using this approach, we pooled the prevalence rate of (1) all the studies that based the diagnosis on the DSM criteria; and then split these into (1a) those in which the diagnosis was established in a face-to-face interview and (1b) those that were established using self-made DSM questions. Furthermore (2) we also calculated the prevalence estimates using the different questionnaires as well as the different cut-offs. Some studies used more than one prevalence rate using different cut-offs (for the Insomnia Severity Scale and the Athens Insomnia Scale). In those cases, we included the studies in multiple analyses. Heterogeneity variance components in our model were estimated using restricted maximum likelihood (REML). Lastly, a 95% prediction interval was computed for each pooled prevalence estimate.

We performed subgroup analyses on the sample of studies in which DSM criteria were used to establish the diagnosis (either in an interview or using self-made questions). Subgroup analyses were conducted for (1) different geographical areas (Europe, North America, South America, Asia, Africa, Middle East, and Oceania), and (2) studies with either low or high study quality. Differences in Insomnia prevalence were statistically tested using likelihood ratio tests where appropriate. Finally, we performed meta-regression analysis to examine the effect of the mean age of the sample and the percentage of males.

3 | Results

3.1 | Selection of Studies

Through our literature search we identified 9455 references and another 7 through manuscript references (Figure 1). After removing duplicates, 6732 manuscripts remained. We excluded 6111 of these based on title and abstract. The remaining 621 manuscripts were retrieved for full-text screening and evaluated by 2 independent assessors. A total of 574 manuscripts were excluded, mostly because they did not measure insomnia appropriately. Ultimately, 47 studies were included in the study.

3.2 | Characteristics of Included Studies

The included studies were published between 2004 and 2024 (Table 1). The sample size of the studies varied between 112

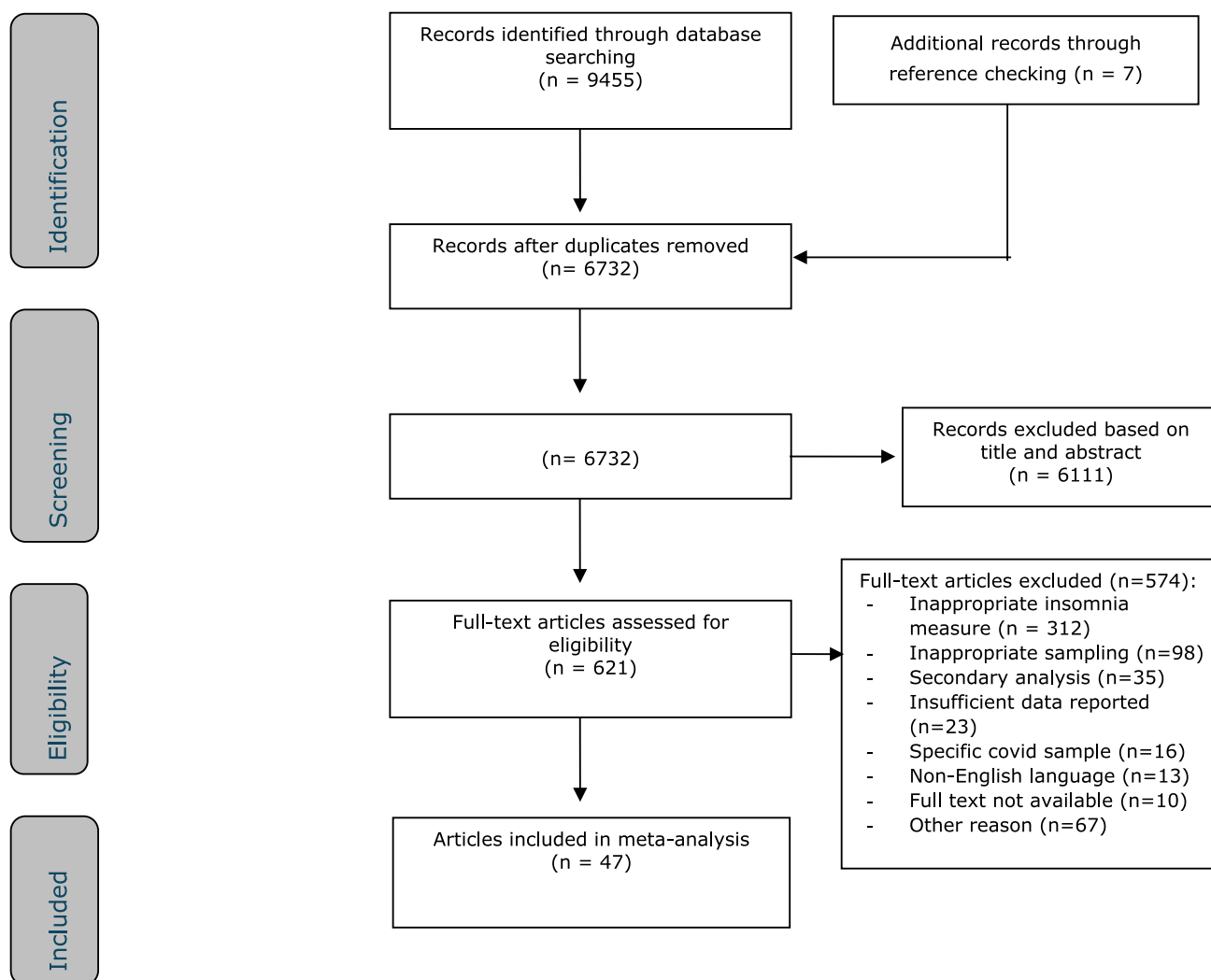


FIGURE 1 | Flowchart of the search and selection procedure of studies.

(Sakamoto et al. 2017) and 20,486 (Hagen 2019). One study included two different samples (Sheaves et al. 2016) resulting in 48 samples. Of these 48 samples, 14 (29.2%) based the insomnia diagnosis on a DSM interview, 10 (20.8%) used self-developed questions to determine the insomnia diagnosis, and the remaining 24 samples (50.0%) used questionnaires (12 used the ISI, 11 the AIS and one the HSDQ). Three of these 24 samples used two different cut-offs to report the prevalence rate (Kim et al. 2016; Luciano et al. 2024; Miner et al. 2018).

Some studies used specific age samples: one included only respondents aged 38 years (Goldman-Mellor et al. 2014), one included respondents aged 25 to 45 years (Léger et al. 2011), and three focused on elderly (either 60+ or 65+; López-Torres Hidalgo et al. 2012; Ma et al. 2018; Sakamoto et al. 2017). Furthermore, one was carried out in a sample of people living on high altitude (Sakamoto et al. 2017) and one was carried out in Hispanics/Latinos (living in the US; Alcántara et al. 2019).

Of all 47 studies, 14 (29.8%) were performed in Asia, 11 (23.4%) in Europe, 8 (17.0%) in North America, 5 (10.6%) in Oceania, 3 (6.4%) in South America, 2 (4.3%) in Africa and 3 (6.4%) in the Middle East, one study was conducted in multiple countries across geographical areas. The mean age in the studies ranged

from 37.0 years (Al Karaki et al. 2020) to 84.3 years (Miner et al. 2018). The proportion of males in the studies ranged from 27.2% (Hagen et al. 2019) to 57.4% (Ade et al. 2021).

3.3 | Quality Assessments

Out of the 47 studies, 29 (61.7%) scored positive on six or more of the nine quality items and were considered to be of high quality (see Supporting Information). The remaining 18 studies scored less and were considered to be of lower quality. The highest scoring item was item 8 “appropriate statistics” which we considered to be the case for all of the studies. The lowest scoring item was item 5 “Analysis conducted with sufficient coverage of identified sample”. Most of the studies (87%) did not report sufficiently whether they investigated selective attrition or not.

3.4 | Overall Effects

The effects were pooled across different studies depending on the way insomnia was measured (Table 2). First, we pooled all the studies using DSM criteria to establish Insomnia

TABLE 1 | Descriptive characteristics of included studies.

Study	Diagnosis	Instrument	Cutoff	Country	Mean age	n sample	% insomnia	% males	% men insomnia	% women insomnia	Quality rating
Adams et al. 2017	DSM SR			Australia		1011	20.0	49.8	16.7	23.2	Low
Ade et al. 2021	Questionnaire	ISI	8	Benin		930	21.6	57.4	16.9	28.0	High
Aernout et al. 2021	DSM F2F	MINI		Multi	44.5	57,298	11.3	46.9	9.9	12.5	Low
Al Karaki et al. 2020	Questionnaire	AIS	6	Lebanon	37.0	756	47.1	36.0			High
Alcántara et al. 2019	Questionnaire	ISI	15	USA		1192	14.9				Low
Ali et al. 2019	Questionnaire	AIS	6	Ethiopia		840	42.9	40.1	33.8	48.9	High
Appleton et al. 2022	DSM SR			Australia		1758	31.7	48.0	28.9	34.3	Low
Bartlett et al. 2008	Questionnaire	AIS	7	Australia		3300	32.5				High
Budhiraja et al. 2011	DSM F2F			USA	41.7	3282	21.4	49.2	18.0	24.7	Low
Calem et al. 2012	DSM F2F	CIS-R		England	38.9	20,503	5.0	45.5			High
Castro et al. 2013	DSM F2F			Brazil		1101	14.7	46.2	9.2	19.9	Low
Chami et al. 2019	DSM F2F			Lebanon	45.4	501	34.5	35.7	24.0	40.4	Low
Chung et al. 2018	DSM F2F	BIQ		China	54.4	398	14.6	38.7			High
de Entrambasaguas et al. 2023	DSM F2F			Spain		2115	13.7	47.8	13.8	13.5	Low
El-Gilany et al. 2018	Questionnaire	AIS	6	Egypt	67.3	474	37.8	49.8			High
Goldman-Mellor et al. 2014	DSM SR			New Zealand	38.0	949	19.6	50.3	15.9	23.3	Low
Hagen et al. 2019	DSM SR			Norway	57.2	20,486	8.1	27.2			High
Han et al. 2017	Questionnaire	AIS	6	China	42.2	8371	12.8	51.5			Low
Hyland et al. 2022	DSM SR	SCI		Ireland		1110	15.0	48.0			High
Jansson-Fröjmark and Linton 2008	DSM SR			Sweden	42.0	1746	9.7	47.0			Low
Kerkhof 2017	Questionnaire	HSDQ	Nr	Netherlands		2089	8.2				High
Khan et al. 2018	Questionnaire	ISI	8	India	39.4	1700	10.3	51.2	11.3	9.3	High
Kim et al. 2016	Questionnaire	ISI	15	South Korea		2695	3.6	49.9	2.5	6.1	High
Kim et al. 2016	Questionnaire	ISI	8	South Korea		2695	12.4	49.9			High

(Continues)

TABLE 1 | (Continued)

Study	Diagnosis	Instrument	Cutoff	Country	Mean age	n sample	% insomnia	% males	% men insomnia	% women insomnia	Quality rating
La et al. 2020	Questionnaire	ISI	10	South Korea		2695	10.7	49.9			High
Lee et al. 2024	Questionnaire	ISI	10	South Korea		3030	13.3	51.2	12.0	14.7	High
Léger et al. 2011	DSM SR	SDQ		France		1004	12.1	49.4	10.1	14	High
López-Torres Hidalgo et al. 2012	DSM F2F			Spain	74.4	926	8.6	45.8	6.4	10.6	Low
Luciano et al. 2024	Questionnaire	ISI	15	Brazil		890	16.7				Low
Luciano et al. 2024	Questionnaire	ISI	8	Brazil		890	45.2				Low
Ma et al. 2018	Questionnaire	AIS	7	China	69.7	3045	23.9	54.9	28.1	18.7	High
Mao et al. 2024	Questionnaire	ISI	8	China		12,544	9.6	53.4	8.3	11.2	High
Matsuura et al. 2020	Questionnaire	AIS	6	Japan		1997	23.6	47.1	21.8	25.2	High
Miner et al. 2018	Questionnaire	ISI	8	USA	84.3	379	43.0	32.2	41.0	44.0	High
Miner et al. 2018	Questionnaire	ISI	15	USA	84.3	379	10.3	32.2			High
Monterrosa-Castro et al. 2013	Questionnaire	AIS	6	Colombia		1325	27.5			27.5	Low
Morin et al. 2006	DSM F2F			Canada	44.7	2001	9.5	42.2	7.9	11.0	High
Novak et al. 2004	Questionnaire	AIS	10	Hungary	48.0	12,653	9.2	44.8	5.5	11.1	High
Potvin et al. 2012	DSM F2F			Canada	73.8	2414	8.8				High
Roth et al. 2011	DSM F2F	BIQ		USA		10,094	22.1				High
Sakamoto et al. 2017	Questionnaire	ISI	8	India	69.3	112	15.2	42.0			Low
Sheaves et al. 2016 (sample 1)	DSM F2F	CIS-R		UK	43.9	8580	7.3				High
Sheaves et al. 2016 (sample 2)	DSM F2F	CIS-R		UK	46.4	7403	8.6				High
Simonelli et al. 2017	Questionnaire	ISI	15	USA	47.0	2156	15.5	35.3			Low
Sivertsen et al. 2009	DSM SR			Norway	49.1	47,700	13.5	46.4	11.2	15.6	High
Sun et al. 2022	Questionnaire	AIS	7	China	50.9	21,376	11.4	46.0	7.7	14.5	High
Sweetman et al. 2021	DSM SR			Australia	46.6	2044	13.3	48.6	14.9	11.8	High

(Continues)

TABLE 1 | (Continued)

Study	Diagnosis	Instrument	Cutoff	Country	Mean age	n sample	% insomnia	% males	% men insomnia	% women insomnia	Quality rating
Taylor et al. 2007	DSM SR			USA		538	25.5				Low
Xie et al. 2021	Questionnaire	AIS	6	China		1213	50.3	45.9	53.1	47.9	High
Yang et al. 2021	Questionnaire	ISI	8	China	70.2	871	45.7	50.1	42.4	49	High
Zailinawati et al. 2008	DSM F2F			Malaysia	49.0	1611	9.6	52.9			Low

Abbreviations: AIS = Athens insomnia scale, CIS-R = revised clinical interview schedule, HSDQ = holland sleep disorder questionnaire, ISI = insomnia severity index, MINI = mini international neuropsychiatric interview, SDQ = sleep condition indicator, USA = United States of America.

Disorder. The overall prevalence was 13.9% (95% CI: 10.9%–17.6%; Figure 2). The prevalence rate of studies assessing the diagnoses through an interviewer or clinician was 12.4% (95% CI: 9.0%–16.8%) while the prevalence rate of studies assessing the diagnoses through self-report was 16.3% (95% CI: 11.3%–23.0%).

We then pooled the prevalence rates across studies using questionnaires with different cut-off scores. For the Athens Insomnia Scale (AIS) the recommended cut-off of 6 yielded a prevalence rate of 32.0% (95% CI: 22.1%–44.6%). Higher AIS cut-off rates (meaning more severe insomnia) yielded lower prevalence rates: 21% for the AIS with a cut-off of 7 (95% CI: 11.0%–36.5%) and 9.0% for a cut-off of 10 (95% CI: 2.6%–27.0%). For the Insomnia Severity Index (ISI) the recommended cut-off of 10 yielded a prevalence rate of 12.5% (95% CI: 5.3%–26.8%). A lower cut-off (ISI 8, meaning less severe insomnia) increased the prevalence to 25.1% (95% CI: 17.8%–34.2%) while a higher cut-off (ISI 15) decreased the prevalence to 7.5% (95% CI: 4.8%–11.4%). There was only one study using the HSDQ (prevalence 8.2%; 95% CI: 2.3%–25.0%).

The prediction intervals (see Table 2), which estimate the range in which the prevalence rate will fall in future studies, were wide for all pooled prevalence rates for example, between 3.5% and 35.5% for the prevalence rate using DSM diagnosis based on an interview.

3.5 | Association Between Prevalence Rate and Geographical Area, Study Quality, Age and Gender

We also examined variables that might be associated with the prevalence rate. For these analyses, we only used the 24 studies in which the prevalence rate was based on the DSM diagnosis (either assessed face-to-face or by self-report).

First, we examined the geographical area in which the study was conducted (Table 3). Geographical area was statistically significantly associated with the prevalence rate ($p < 0.01$). Most studies ($n = 10$) were conducted in Europe, with a prevalence rate of 9.9% (95% CI: 6.4%–20.2%). The prevalence rate of North America ($n = 5$) was 16.2% (95% CI: 11.4%–22.6%) and for Oceania ($n = 4$) the prevalence rate was 19.2% (95% CI: 2.8%–27.8%). For the other 3 areas only a few studies were available.

Of the 24 studies using the DSM, half (50%) were considered to be of high quality while the remaining studies were considered to be of lower quality either because they did not adhere to the quality criteria or because this was not clearly reported. The difference in quality was significantly associated with the prevalence rate ($p = 0.02$). High quality studies reported lower prevalence rates (10.7%) than lower quality studies (17.1%).

The mean age of the study samples was not associated with the prevalence rate ($p = 0.98$) nor was the percentage of males in the studies ($p = 0.94$; Table 4). However, quite a few studies did not report mean age of the sample but used age categories instead ($n = 21$; 44.6%), and some studies did not report the crude number or proportion of (fe)males ($n = 6$; 12.8%).

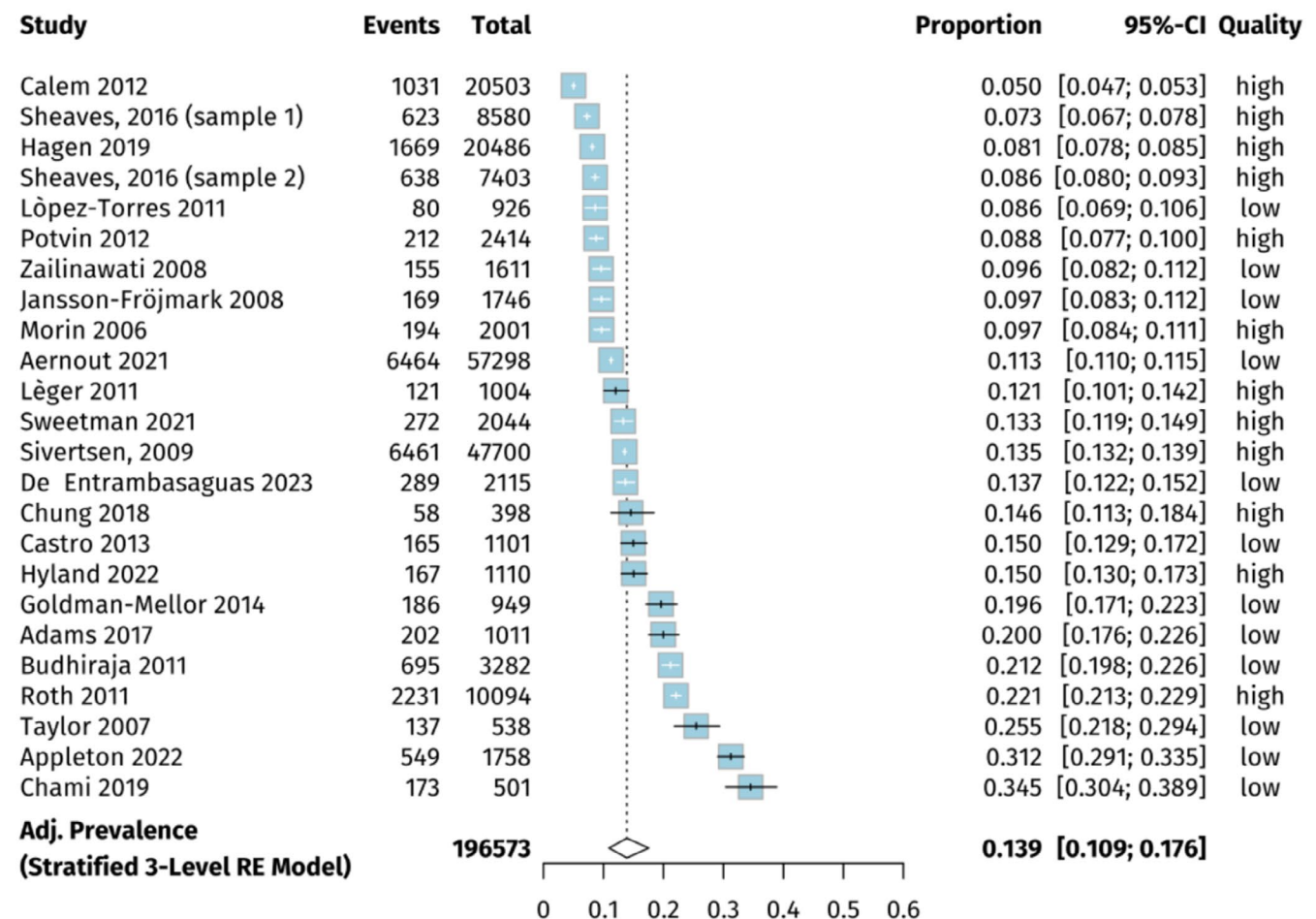
TABLE 2 | Overall results (stratified three-level model).

Instrument type	n_{est}	Prevalence	Lower CI	Upper CI	Lower PI	Upper PI
Diagnosis (overall) ^a	24	13.9%	10.9%	17.6%	4.0%	38.3%
Diagnosis (F2F)	14	12.4%	9.0%	16.8%	3.5%	35.5%
Diagnosis (self-report)	10	16.3%	11.3%	23.0%	4.7%	43.6%
Questionnaire (AIS-6)	7	32.3%	22.1%	44.6%	10.4%	66.2%
Questionnaire (AIS-7)	3	21.0%	11.0%	36.5%	5.5%	54.9%
Questionnaire (AIS-10)	1	9.0%	2.6%	27.0%	1.5%	38.8%
Questionnaire (ISI-8)	8	25.1%	17.8%	34.2%	7.8%	57.2%
Questionnaire (ISI-10)	2	12.5%	5.3%	26.8%	2.8%	41.7%
Questionnaire (ISI-15)	5	7.5%	4.8%	11.4%	2.0%	24.6%
Questionnaire (HSDQ)	1	8.2%	2.3%	25.0%	1.4%	36.4%

Note: $I^2 = 99.60\%$, $\tau^2_{\text{between}} = 0.422$; $\tau^2_{\text{within}} = 0.024$.

Abbreviations: CI = confidence interval, n_{est} = number of prevalence estimates, PI = prediction interval.

^aFitted in a separate model.

**FIGURE 2** | Forest plot of the prevalence of 24 studies using DSM criteria to assess the insomnia diagnosis.

4 | Discussion

The results of this meta-analysis provide a comprehensive overview of the global prevalence of insomnia and highlight several important findings. As expected, the prevalence rate

varied depending on the method used to assess insomnia. In our opinion, the DSM diagnosis established by an interviewer or clinician is the gold standard. The pooled prevalence rate of the 14 studies using this method was 12.4% (mostly based on DSM-IV criteria, 64%). Because prevalence studies often include

TABLE 3 | Subgroup analyses for geographical area and study quality (based on 24 studies using DSM diagnosis).

Subgroup	n_{est}	Prevalence	Lower CI	Upper CI	Lower PI	Upper PI	I^2	τ^2_{between}	τ^2_{within}	χ^2	p
Country											
Asia	2	11.6%	6.4%	20.2%	4.2%	28.5%	99.08%	0.104	0.104	19.124	0.002
Europe	10	9.9%	7.6%	12.7%	4.1%	21.9%					
Middle East	1	39.5%	19.1%	64.3%	14.4%	71.6%					
North America	5	16.2%	11.4%	22.6%	6.8%	34.1%					
Oceania	4	19.2%	12.8%	27.8%	7.9%	39.6%					
South America	1	14.6%	6.4%	29.9%	4.6%	38.1%					
Study quality											
High	12	10.7%	8.0%	14.1%	3.8%	26.5%	99.51%	0.145	0.145	5.649	0.017
Low	12	17.1%	13.1%	22.1%	6.4%	38.3%					

Abbreviation: n_{est} = number of prevalence estimates.

large samples, some authors preferred to use self-reported diagnosis based on questions that aim to assess the DSM criteria of Insomnia Disorder. These studies reported a higher prevalence rate (16.3%). It must be noted however that it was not always easy to make a distinction between these two different ways of establishing the diagnosis. For example, it was difficult to make a distinction between an interviewer who had been trained to perform a 'true' diagnostic interview and between an interviewer just asking a number of predetermined questions. There were also quite a few studies that used questionnaires which are validated against the DSM diagnosis (mostly AIS and ISI). Interestingly, they did not always use the reported optimal cut-off score. For the AIS, the recommended cut-off of 6, and the pooled prevalence of the 7 studies that used this cut-off is 32.0%. The pooled prevalence rate of the 8 studies that used the ISI with the recommended cut-off of 10 was 12.5%, remarkably similar to the rate of the studies using clinician-based DSM criteria. However, it must be noted that all the prevalence rates had large confidence and prediction intervals, indicating considerable uncertainty around these rates.

More than 20 years ago Ohayon published a seminal overview of the epidemiology of insomnia (Ohayon 2002). He reported prevalence rates for insomnia symptoms with daytime consequences as well as for DSM-IV diagnosis of insomnia. The rates were between 9% and 15% for the first category and 6% for the latter. Similar numbers were observed by Morin and Jarrin (2022) with a 6%–10% prevalence rate using stringent criteria. Our prevalence rate based on DSM criteria is considerably higher than those reported before. This might be due to a real increase of insomnia over time but may also be affected by differences between studies. Ohayon already noted in 2002 that insomnia is evaluated idiosyncratically in epidemiological studies, and it seems that 20 years later, this is still the case. In the absence of a clear definition on insomnia, and consensus on how it should be measured, there is large variety in the way it is assessed.

This study showed that the prevalence rate was associated with the different regions around the world. Unfortunately, there were very few studies for Asia, Middle East and South America making comparisons to those parts of the world difficult. There was considerable variation in the prevalence rates for the remaining parts of the world: Europe 9.9%, North America 16.2% and Oceania 19.2%. One of the included studies (Aernout et al. 2021) included participants from different countries around the world and also observed markedly different prevalence rates, as did a study which looked into prevalence rates of insomnia around the world among people with sleep apnea (Zhang et al. 2019). These regional differences could be attributed to various socioeconomic and cultural factors. For example, there might be differences in urbanisation, exposure to noise and exposure to (sun)light (Brockmann et al. 2017). Furthermore, all questionnaires were developed in western countries and not well validated for other parts of the world. Further research is needed to explore these regional disparities and identify underlying causes.

Unexpectedly, there was no association between gender and the prevalence of insomnia, nor between age and the prevalence of insomnia. The effect of age and gender have been firmly established in previous research, with older people and females

TABLE 4 | Moderator analyses for mean age and gender.

Moderator	n_{est}	b	S.E.	Z	Lower CI	Upper CI	p	I^2	τ^2_{between}	τ^2_{within}
Mean age	16	-0.122	0.561	-0.218	0.827	-1.222	0.978	99.98%	2.359	2.359
% males	19	0.036	0.463	0.078	0.938	-0.871	0.943	99.96%	1.925	1.925

Abbreviation: n_{est} = number of prevalence estimates.

reporting insomnia more often than younger people and males (Ohayon 2002; Riemann et al. 2022). The reason for this might be that we only looked at the mean age of the sample while the distribution of age might have been skewed due to the inclusion of some very young or very old respondents. Many studies (44.6%) did not report the mean age but instead reported the percentages of respondents falling into different age categories (without reporting the prevalence rates for those categories). For gender we also only looked at the percentage of males in the sample and did not pool the prevalence rates separately because these were often unavailable. Table 1 shows, however, that 12 of the 14 studies using DSM criteria showed higher prevalence rates for females than for males. In conclusion, we only can say that there was insufficient data to draw firm conclusions about the patient-level predictions of age or gender.

Another limitation of this meta-analysis is the quality of the included studies even though 61.7% of the studies was scored as high quality. As can be expected, high-quality studies reported a lower prevalence (10.7%), compared to lower-quality studies (17.1%). The studies with lower quality often (45%) did not report on the sample size they had set out to reach. They also often (87%) did not report whether there was selective response to the survey. All this might mean that our overall prevalence rate of 12.4% might be lower in reality.

We believe that we are among the first to meta-analytically review all the reported prevalence rates of insomnia around the world. The strength of our study is that we only included studies which used true random samples of the general population even though we allowed true random samples of populations based on a specific age, gender or ethnicity. We also only included studies that encompassed the DSM criteria (including daytime impairments) or questionnaires that are validated against DSM criteria.

Based on our findings several points should be considered when determining the prevalence of insomnia in a valid way. First, we recommend using DSM-5 criteria to establish the diagnosis, which should be assessed with a standard and validated instrument that can be used by professionals as well as by trained lay persons. However, we recognise it is often not feasible to perform diagnostic interviews due to high costs and patient burden. In these cases an alternative is to use a standard questionnaire which has been validated against DSM criteria, preferably the DSM-5. We recommend using one of two well validated options. The first is the Sleep Condition Indicator (SCI; Espie et al. 2014). This is a short 8 item instrument assessing the DSM-5 criteria. The SCI is valid and reliable (Hellström et al. 2019; Wong et al. 2017). The second one is the ISI, a questionnaire validated against the DSM-IV rather than the DSM-5, but one that is widely used in insomnia research, which enhances comparability (Morin et al. 2011). A strong feature is that the online

version is also validated (Thorndike et al. 2011). For now, we recommend using the validated cut-off of 10 points based on the DSM-IV criteria, but validating against the latest version of a DSM-5 clinical interview is essential. Based on the results of our meta-analyses, we do not recommend using the AIS since the standard cut-off score of 6 yielded a much higher prevalence rate than the other instruments.

5 | Conclusion

This meta-analysis highlights the significant global burden of insomnia. In our view the best estimate of the prevalence, based on DSM interviews, is 12.4%. However, this meta-analysis also showed the need for standardised ways of assessing the insomnia diagnoses. The confidence and prediction intervals around the pooled estimates are wide indicating that there is substantial uncertainty. We strongly recommend the sleep research community to use the golden standard of a standardised clinical interview as much as possible. If this is not feasible, the two instruments we currently recommend are the Sleep Condition Indicator, which is based on DSM-5 criteria, or the Insomnia Severity Index with a cut-off of 10, which although based on the DSM-IV resulted in a prevalence estimate close to that of the clinical interviews. Future research should address regional disparities and understand other factors contributing to the variability in prevalence rates. This will be crucial in developing effective interventions and public health strategies to manage and reduce the impact of insomnia worldwide.

Author Contributions

Annemieke van Straten: conceptualization, writing – original draft, investigation, supervision, project administration. **Karl Juri Weinreich:** writing – review and editing, validation, data curation. **Bernát Fábíán:** validation, writing – review and editing, data curation. **Joyce Reesen:** validation, writing – review and editing, data curation. **Sarah Grigori:** validation, writing – review and editing, data curation. **Annemarie I. Luik:** methodology, writing – review and editing. **Mathias Harrer:** methodology, writing – review and editing, formal analysis. **Jaap Lancee:** conceptualization, writing – review and editing, supervision.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.